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COMPUTER NETWORKS

Ex.No.1	**Study of various Network commands used in Linux and Windows: Hands-on practice of various network commands.**
Date:	

AIM: - To Study of various Network commands used in Linux and Windows

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	`arp -a`	`arp -a`	Displays the ARP cache, showing the IP addresses and corresponding MAC addresses of devices on the local network.	`arp -a`	`Interface: 192.168.1.10 --- 0x7 Internet Address Physical Address Type 192.168.1.1 00-1a-2b-3c-4d-5e dynamic`
2	`hostname`	`hostname`	Displays the hostname (computer name) of the local system.	`hostname`	`DESKTOP-PC01`
3	`ipconfig /all`	`ipconfig /all`	Displays complete TCP/IP configuration, including IP address, subnet mask, gateway, DNS, DHCP, and MAC address.	`ipconfig /all`	`IPv4 Address : 192.168.1.10 Default Gateway : 192.168.1.1 DNS Server : 8.8.8.8 DHCP Enabled : Yes`
4	`nbtstat -a`	`nbtstat -a <hostname>`	Displays NetBIOS name table of a remote computer for troubleshooting NetBIOS over TCP/IP.	`nbtstat -a DESKTOP-PC01`	`NetBIOS Remote Machine Name Table Name Type Status DESKTOP-PC01 <00> UNIQUE Registered`
5	`netstat`	`netstat`	Displays active TCP/IP connections, listening ports,	`netstat`	`Proto Local Address Foreign Address State TCP 192.168.1.10:49762`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			routing tables, and interface statistics.		142.250.190.78:443 ESTABLISHED
6	netstat -r	netstat -r	Displays the IP routing table of the local computer.	netstat -r	IPv4 Route Table Network Destination Netmask Gateway Interface
7	nslookup	nslookup <domain>	Queries the DNS server to obtain the IP address or DNS records of a domain.	nslookup www.google.com	Server: dns.google Address: 8.8.8.8 Name: www.google.com Address: 142.250.190.68
8	pathping	pathping <hostname/IP>	Combines the features of ping and tracert to trace the route and measure packet loss at each hop.	pathping www.google.com	Tracing route to www.google.com Computing statistics... Hop 1 192.168.1.1 Hop 2 10.0.0.1 Packet Loss: 0%
9	ping	ping <hostname/IP>	Tests connectivity between the local computer and another host using ICMP Echo Request and Reply messages.	ping www.facebook.com	Pinging www.facebook.com [157.240.22.35] with 32 bytes of data: Reply from 157.240.22.35: bytes=32 time=25ms TTL=54
10	route	route print	Displays or modifies the IP routing table. It is commonly used to view or configure static routes.	route print	IPv4 Route Table Active Routes: Network Destination Netmask Gateway Interface

Some important Linux networking commands

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	ip	`ip <OPTIONS> <OBJECT> <COMMAND>`	Displays and manages IP addresses, routing tables, and network interfaces.	`ip address show`	`2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0`
2	ip address add	`ip address add <IP>/<Prefix> dev <Interface>`	Assigns an IP address to a network interface.	`ip address add 192.168.1.254/24 dev eth0`	`Assigned 192.168.1.254/24 to eth0 successfully.`
3	ip address del	`ip address del <IP>/<Prefix> dev <Interface>`	Removes an IP address from a network interface.	`ip address del 192.168.1.254/24 dev eth0`	`Deleted 192.168.1.254/24 from eth0.`
4	ip link set up	`ip link set <Interface> up`	Brings the network interface online.	`ip link set eth0 up`	`Interface eth0 is UP.`
5	ip link set down	`ip link set <Interface> down`	Brings the network interface offline.	`ip link set eth0 down`	`Interface eth0 is DOWN.`
6	ip link set promisc on	`ip link set <Interface> promisc on`	Enables promiscuous mode on the interface.	`ip link set eth0 promisc on`	`Promiscuous mode enabled on eth0.`
7	ip route add default	`ip route add default via <Gateway> dev <Interface>`	Adds the default gateway for all outgoing traffic.	`ip route add default via 192.168.1.254 dev eth0`	`Default route added successfully.`
8	ip route add	`ip route add <Network> via <Gateway>`	Adds a static route through a gateway.	`ip route add 192.168.1.0/24 via 192.168.1.254`	`Route added successfully.`
9	ip route add (Direct Route)	`ip route add <Network> dev <Interface>`	Adds a direct route through a network interface.	`ip route add 192.168.1.0/24 dev eth0`	`Route added via eth0.`
10	ip route delete	`ip route delete <Network> via <Gateway>`	Deletes a static route.	`ip route delete 192.168.1.0/24`	`Route deleted successfully.`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
				via 192.168.1.254	
11	ip route get	ip route get <Destination IP>	Displays the route used to reach a destination.	ip route get 10.10.1.4	10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100
12	ifconfig	ifconfig	Displays and configures network interface parameters (deprecated in modern Linux).	ifconfig	eth0: flags=4163<UP,BROADCAST,RUNNING> inet 192.168.1.100 netmask 255.255.255.0
13	mtr	mtr <hostname/IP>	Combines Ping and Traceroute to analyze network paths.	mtr google.com	HOST: localhost Loss% Snt Last Avg Best Wrst StDev
14	mtr -g	mtr -g <hostname>	Displays numeric IP addresses instead of hostnames.	mtr -g google.com	192.168.1.1 → 142.250.190.78
15	mtr -b	mtr -b <hostname>	Displays both hostnames and IP addresses.	mtr -b google.com	router.local (192.168.1.1) google.com (142.250.190.78)
16	mtr -c	mtr -c <count> <hostname>	Sends a specified number of probes.	mtr -c 10 google.com	10 probes completed successfully.
17	tcpdump -D	tcpdump -D	Lists all available network interfaces.	tcpdump -D	1.eth0 2.lo 3.wlan0
18	tcpdump	tcpdump -i <Interface>	Captures packets on a network interface.	tcpdump -i eth0	Listening on eth0...
19	tcpdump -c	tcpdump -i <Interface> -c <count>	Captures a specified	tcpdump -i eth0 -c 10	10 packets captured.

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			number of packets.		
20	tcpdump host	`tcpdump -i <Interface> host <IP>`	Captures traffic to and from a specific host.	`tcpdump -i eth0 host 8.8.8.8`	`Captured packets to/from 8.8.8.8.`
21	tcpdump src host	`tcpdump -i <Interface> src host <IP>`	Captures packets originating from a specific host.	`tcpdump -i eth0 src host 8.8.8.8`	`Source packets from 8.8.8.8 displayed.`
22	tcpdump dst host	`tcpdump -i <Interface> dst host <IP>`	Captures packets destined for a specific host.	`tcpdump -i eth0 dst host 8.8.8.8`	`Destination packets to 8.8.8.8 displayed.`
23	tcpdump net	`tcpdump -i <Interface> net <Network>/<Prefix>`	Captures traffic from a specified network.	`tcpdump -i eth0 net 10.1.0.0/24`	`Packets from network 10.1.0.0/24 displayed.`
24	tcpdump port	`tcpdump -i <Interface> port <Port>`	Captures traffic on a specific port.	`tcpdump -i eth0 port 53`	`DNS packets captured.`
25	tcpdump host and port	`tcpdump -i <Interface> host <IP> and port <Port>`	Captures traffic for a specific host and port.	`tcpdump -i eth0 host 8.8.8.8 and port 53`	`DNS traffic with host 8.8.8.8 displayed.`
26	tcpdump port not	`tcpdump -i <Interface> port not <Port1> and not <Port2>`	Captures all traffic except the specified ports.	`tcpdump -i eth0 port not 53 and not 25`	`Traffic excluding ports 53 and 25 captured.`
27	ping	`ping <hostname/IP>`	Tests network connectivity using ICMP Echo Requests and Echo Replies.	`ping google.com`	`64 bytes from google.com: icmp_seq=1 ttl=117 time=10.5 ms`
28	ping -c	`ping -c <count> <hostname>`	Sends a specified number of	`ping -c 10 google.com`	`10 packets transmitted, 10 received, 0% packet loss.`

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			ICMP Echo Requests.		

Configuring an Ethernet connection by using nmcli:

Step No.	Task	Command	Expected Result
1	List all Network Manager connection profiles	`nmcli connection show`	Displays all available network connection profiles with their **Name, UUID, Type, and Device**.
2	Create a new Ethernet connection profile It is specifically an **`nmcli` connection management command**. • **`nmcli`** → NetworkManager Command Line Interface • **`connection`** → Manages network connection profiles. • **`add`** → Creates a new connection profile.	`nmcli connection add con-name Internal-LAN ifname enp1s0 type ethernet`	Creates a new Ethernet connection profile named **Internal-LAN**.
3	Rename an existing connection profile (Optional)	`nmcli connection modify "Wired connection 1" connection.id Internal-LAN`	Renames the existing connection profile to **Internal-LAN**.
4	Display the current settings of the connection profile	`nmcli connection show "Internal-LAN"`	Displays interface name, autoconnect status, IPv4 and IPv6 configuration details.
5	Configure IPv4 using DHCP	`nmcli connection modify "Internal-LAN" ipv4.method auto`	Configures the interface to obtain the IPv4 address

Step No.	Task	Command	Expected Result
			automatically using DHCP.
6	Configure a static IPv4 address	`nmcli connection modify "Internal-LAN" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com`	Assigns a static IPv4 address, gateway, DNS server, and DNS search domain.
7	Configure IPv6 using SLAAC	`nmcli connection modify "Internal-LAN" ipv6.method auto`	Enables automatic IPv6 configuration using Stateless Address Autoconfiguration (SLAAC).
8	Configure a static IPv6 address	`nmcli connection modify "Internal-LAN" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::ffee ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com`	Assigns a static IPv6 address, gateway, DNS server, and search domain.
9	Activate the connection profile	`nmcli connection up Internal-LAN`	Activates the **Internal-LAN** network connection profile.
10	Verify the IP address configuration	`ip address show enp1s0`	Displays the assigned IPv4 and IPv6 addresses of the **enp1s0** network interface.
11	Verify the IPv4 default gateway	`ip route show default`	Displays the configured IPv4 default gateway.
12	Verify the IPv6 default gateway	`ip -6 route show default`	Displays the configured IPv6 default gateway.
13	Verify the DNS configuration	`cat /etc/resolv.conf`	Displays the configured DNS servers and DNS search domain.
14	Test network connectivity	`ping <host-name-or-IP-address>`	Sends ICMP Echo Requests to verify connectivity with the specified host.

Sample Verification Outputs

Verification Command	Sample Output
ip address show enp1s0	inet 192.0.2.1/24 inet6 2001:db8:1::fffe/64
ip route show default	default via 192.0.2.254 dev enp1s0 proto static
ip -6 route show default	default via 2001:db8:1::ffee dev enp1s0 proto static
cat /etc/resolv.conf	search example.com nameserver 192.0.2.200 nameserver 2001:db8:1::ffbb
ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=11.2 ms

Output:

```
student@fedora:~$ arp -a
_gateway (10.0.2.2) at 52:55:0a:00:02:02 [ether] on enp0s3
student@fedora:~127$ hostname
fedora
```

```
student@fedora:~127$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 fedora:bootpc           _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     22939
unix  3      [ ]         STREAM     CONNECTED     22757    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22589    /run/user/1000/at-spi/bus_0
unix  2      [ ]         DGRAM                    18681    /run/systemd/home/notify
unix  3      [ ]         STREAM     CONNECTED     19513    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23704
unix  3      [ ]         STREAM     CONNECTED     23002
unix  3      [ ]         STREAM     CONNECTED     21482    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     21176    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     19398
unix  3      [ ]         DGRAM      CONNECTED     17375
unix  2      [ ]         DGRAM      CONNECTED     16660
unix  3      [ ]         STREAM     CONNECTED     19667
unix  3      [ ]         STREAM     CONNECTED     21249    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     19763    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     22770    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     23282    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     23497    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22269    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     23113    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     22914    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22635    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21191
unix  3      [ ]         STREAM     CONNECTED     23738
unix  3      [ ]         STREAM     CONNECTED     21192
unix  3      [ ]         STREAM     CONNECTED     22680
unix  3      [ ]         STREAM     CONNECTED     22610    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22922
unix  3      [ ]         STREAM     CONNECTED     21199    /run/user/1000/at-spi/bus_0
unix  3      [ ]         STREAM     CONNECTED     23540    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     22761    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     23011
unix  3      [ ]         STREAM     CONNECTED     22887    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     21391    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     23725    @/tmp/.X11-unix/X0
unix  3      [ ]         STREAM     CONNECTED     22489
```

```
student@fedora:~$ netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
default         _gateway        0.0.0.0         UG        0 0          0 enp0s3
```

```
student@fedora:~2$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 380  bytes 52722 (51.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 468  bytes 45472 (44.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 39  bytes 3622 (3.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 39  bytes 3622 (3.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
student@fedora:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.67.46
Name:   google.com
Address: 2404:6800:4007:805::200e
```

```
student@fedora:~127$ ping facebook.com
PING facebook.com (57.144.56.1) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=1 ttl=255 time=14.2 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=2 ttl=255 time=21.2 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=3 ttl=255 time=8.87 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=4 ttl=255 time=16.2 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=5 ttl=255 time=13.4 ms
64 bytes from edge-star-mini-shv-02-maa5.facebook.com (57.144.56.1): icmp_seq=6 ttl=255 time=19.7 ms
^C
--- facebook.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5192ms
rtt min/avg/max/mdev = 8.867/15.598/21.155/4.092 ms
```

```
student@fedora:~$ route print
Usage: route [-nNvee] [-FC] [<AF>]           List kernel routing tables
       route [-v] [-FC] {add|del|flush} ...  Modify routing table for AF.

       route {-h|--help} [<AF>]              Detailed usage syntax for specified AF
       route {-V|--version}                  Display version/author and exit.

        -v, --verbose            be verbose
        -n, --numeric            don't resolve names
        -e, --extend             display other/more information
        -F, --fib                display Forwarding Information Base (default)
        -C, --cache              display routing cache instead of FIB

  <AF>=Use -4, -6, '-A <af>' or '--<af>'; default: inet
  List of possible address families (which support routing):
    inet (DARPA Internet) inet6 (IPv6) ax25 (AMPR AX.25)
    netrom (AMPR NET/ROM) ipx (Novell IPX) ddp (Appletalk DDP)
    x25 (CCITT X.25)
```

```
student@fedora:~3$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 83951sec preferred_lft 83951sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dynamic noprefixroute
       valid_lft 85914sec preferred_lft 13914sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

```
student@fedora:~$ mtr -g google.com
mtr: invalid option -- 'g'

Usage:
 mtr [options] hostname

 -F, --filename FILE        read hostname(s) from a file
 -4                         use IPv4 only
 -6                         use IPv6 only
 -u, --udp                  use UDP instead of ICMP echo
 -T, --tcp                  use TCP instead of ICMP echo
 -I, --interface NAME       use named network interface
 -a, --address ADDRESS      bind the outgoing socket to ADDRESS
 -f, --first-ttl NUMBER     set what TTL to start
 -m, --max-ttl NUMBER       maximum number of hops
 -U, --max-unknown NUMBER   maximum unknown host
 -P, --port PORT            target port number for TCP, SCTP, or UDP
 -L, --localport LOCALPORT  source port number for UDP
 -s, --psize PACKETSIZE     set the packet size used for probing
 -B, --bitpattern NUMBER    set bit pattern to use in payload
 -i, --interval SECONDS     ICMP echo request interval
 -G, --gracetime SECONDS    number of seconds to wait for responses
 -Q, --tos NUMBER           type of service field in IP header
 -e, --mpls                 display information from ICMP extensions
 -Z, --timeout SECONDS      seconds to keep probe sockets open
 -M, --mark MARK            mark each sent packet
 -r, --report               output using report mode
 -w, --report-wide          output wide report
 -c, --report-cycles COUNT  set the number of pings sent
 -j, --json                 output json
 -x, --xml                  output xml
 -C, --csv                  output comma separated values
 -l, --raw                  output raw format
 -p, --split                split output
 -t, --curses               use curses terminal interface
     --displaymode MODE     select initial display mode
 -n, --no-dns               do not resolve host names
 -b, --show-ips             show IP numbers and host names
 -o, --order FIELDS         select output fields
 -y, --ipinfo NUMBER        select IP information in output
 -z, --aslookup             display AS number
 -h, --help                 display this help and exit
 -v, --version              output version information and exit
```

RESULT:

The various network commands in Linux and Windows were studied and executed successfully.